

Machine Learning Assignment

Decision Tree Classification using Comprehensive Exploratory Data Analysis (EDA)

Course: Machine Learning

Dataset: Hotel Booking Cancellation Dataset

Total Marks: 100

Objective

Perform comprehensive EDA, preprocess the Hotel Booking Cancellation Dataset, build and evaluate Decision Tree models, tune hyperparameters, and interpret results.

Dataset

Dataset: <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>

Target Variable: is_canceled

Part A: Data Understanding

- Describe the business problem.
- Explain each feature.
- Identify numerical, categorical, binary and date features.
- Identify the target variable.
- State whether it is classification or regression.

Part B: Exploratory Data Analysis

- Dataset overview (shape, data types, duplicates).
- Missing value analysis and treatment.
- Univariate analysis using histograms, KDE, boxplots and count plots.
- Outlier detection using IQR and Z-score.
- Target variable analysis.
- Correlation analysis, heatmap, scatter plots and categorical analysis.
- Create at least three engineered features.
- Use Chi-square, ANOVA or Mutual Information for feature selection.

Part C: Data Preprocessing

- Handle missing values.
- Encode categorical variables.
- Discuss feature scaling.
- Handle outliers and class imbalance.
- Split data into train/validation/test sets.

Part D: Decision Tree Modeling

- Build a baseline Decision Tree.
- Evaluate using Accuracy, Precision, Recall, F1-score, ROC-AUC and Confusion Matrix.
- Tune hyperparameters using GridSearchCV or RandomizedSearchCV.
- Compare tuned and baseline models.

Part E: Model Interpretation

- Visualize the tree.
- Discuss feature importance.
- Interpret at least five decision rules.

Part F: Model Evaluation

- Compare evaluation metrics and discuss the most appropriate metric.

Part G: Critical Thinking

- Explain interpretability, overfitting, pruning, Gini vs Entropy and comparison.

Deliverables

- Jupyter Notebook (.ipynb)
- Python Script (.py)
- Final Report (PDF)
- Presentation (10 slides)